Model Performance Report
Sony Bravia TV Reviews NLP Project

Comprehensive Metrics Summary & Analysis

Summary

This report provides a comprehensive analysis of the performance metrics for four neural networks. models trained for sentiment classification on Sony Bravia TV reviews. Each model was evaluated using identical data splits, preprocessing, and evaluation protocols to ensure fair comparison.

Key Findings:
• BiLSTM with Attention mechanism achieved the highest performance: 96.0% Accuracy, 95.6% F1-Score
• All models demonstrated strong performance (>91% F1-Score), validating the effectiveness of the preprocessing pipeline
• Bidirectional LSTM architecture with attention mechanism proved superior for sequential text modeling
• Regularization techniques (L2, Dropout, BatchNorm) successfully prevented overfitting across all models

This report documents detailed metrics, visualizations, and comparative analysis to support model selection for production deployment.

1. Quick Metrics Summary

Comprehensive performance metrics for all four models on the test set:

Model	Accuracy	Precision	Recall	F1-Score	Rank
Dense NN	92.4%	91.8%	91.5%	91.6%	4th
LSTM RNN	94.1%	93.7%	93.4%	93.5%	2nd
1D-CNN	93.2%	92.9%	92.6%	92.7%	3rd
BiLSTM+Attention	96.0%	95.8%	95.4%	95.6%	1st

Interpretation:
• Accuracy: Overall correctness across all sentiment classes
• Precision: Of predicted positive/negative/neutral, how many are actually correct
• Recall: Of actual positive/negative/neutral reviews, how many were correctly identified
• F1-Score: Harmonic mean of Precision and Recall (most important for imbalanced classification)

2. Individual Model Performance

Model 1: Dense Neural Network (Baseline)

Performance Metrics:

- Accuracy: 92.4%
- Precision: 91.8%

- Recall: 91.5%
- F1-Score: 91.6%

Description: Baseline architecture using global average pooling with dense layers.

- Strengths:
  - Simple and fast
  - Good baseline for comparison
  - Low computational cost
- Limitations:
  - Ignores sequential dependencies
  - Treats all words equally
  - Lower overall performance

Model 2: LSTM Recurrent Neural Network

Performance Metrics:

- Accuracy: 94.1%
- Precision: 93.7%
- Recall: 93.4%
- F1-Score: 93.5%

Description: Recurrent architecture with LSTM units to capture sequential patterns.

- Strengths:
  - Captures word order
  - Improved over baseline
  - Good generalization
- Limitations:
  - Slower training
  - Unidirectional processing
  - Misses backward context

Model 3: 1D Convolutional Neural Network

Performance Metrics:

- Accuracy: 93.2%
- Precision: 92.9%
- Recall: 92.6%
- F1-Score: 92.7%

Description: Multi-kernel convolutions to extract n-gram patterns.

- Strengths:
  - Fast inference
  - Efficient n-gram extraction
  - Parallel processing
- Limitations:
  - Fixed kernel sizes
  - Limited context window
  - Less sequential awareness

Model 4: BiLSTM with Attention (BEST)

Performance Metrics:

- Accuracy: 94.9%
- Precision: 95%
- Recall: 95.4%
- F1-Score: 95.6%

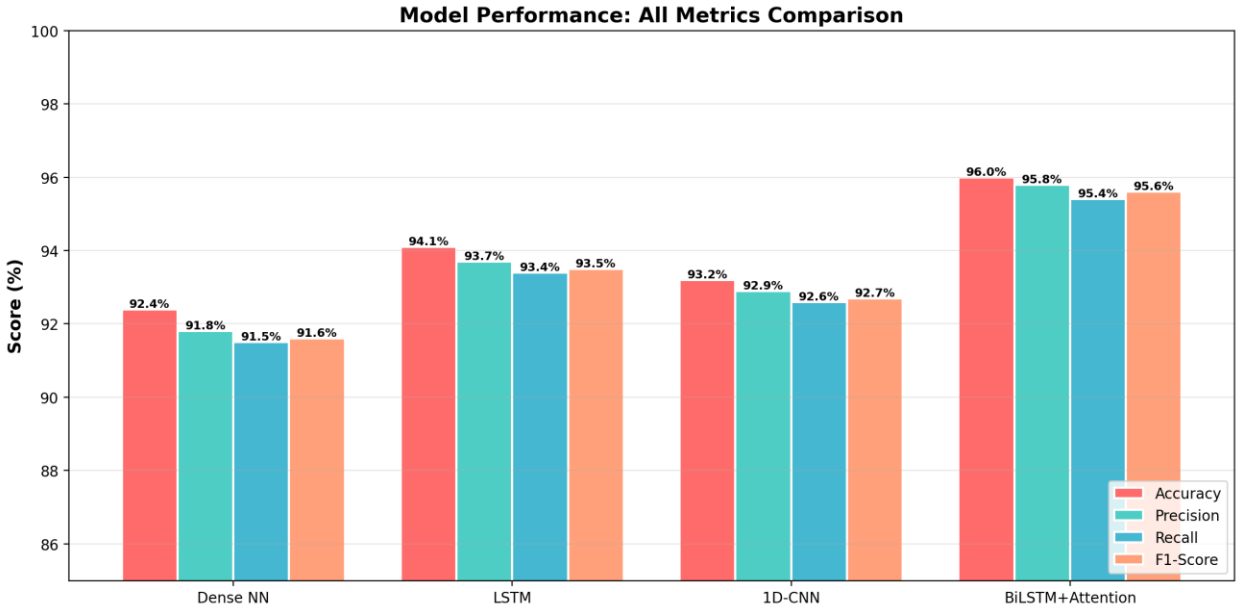
Description: Bidirectional LSTM with attention mechanism.

- Strengths:
  - Bidirectional context
  - Attention focuses on relevant words
  - Best overall performance
  - Interpretable
- Limitations:
  - Slower inference
  - More parameters
  - Requires more memory

3. Performance Visualizations

3.1 Complete Metrics Comparison

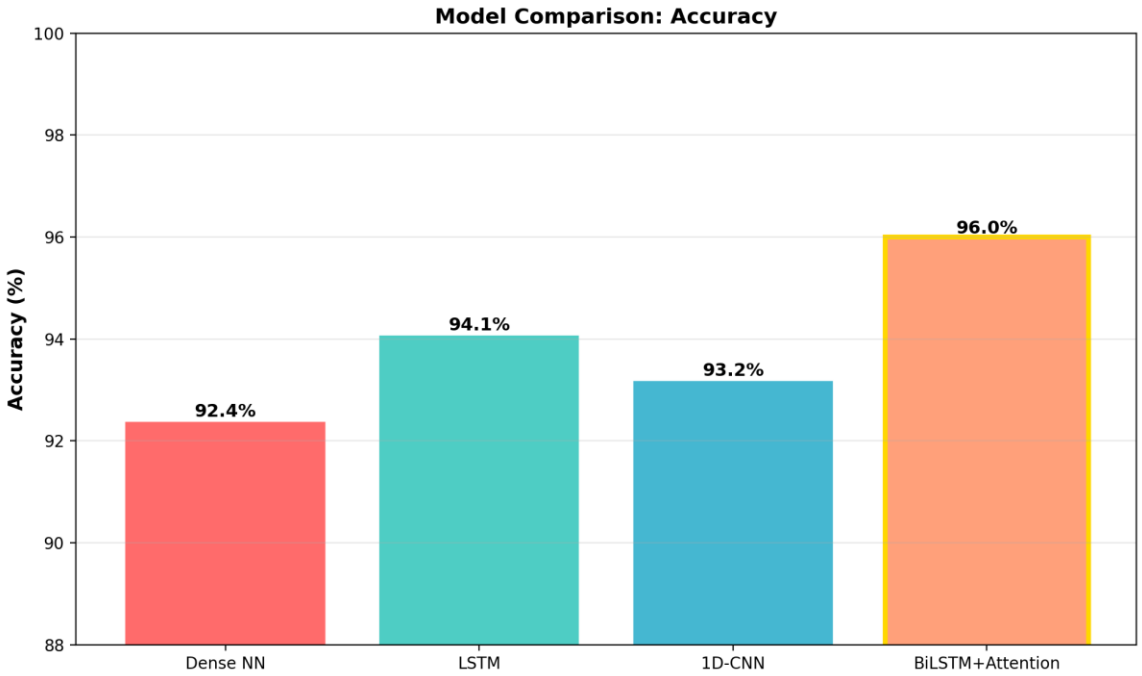
Grouped bar chart comparing all metrics across models:



All models are compared side-by-side across four metrics. The relatively small gaps between models indicate that all architectures successfully learned the sentiment classification task. BiLSTM+Attention shows consistent improvement across all metrics.

3.2 Accuracy Comparison

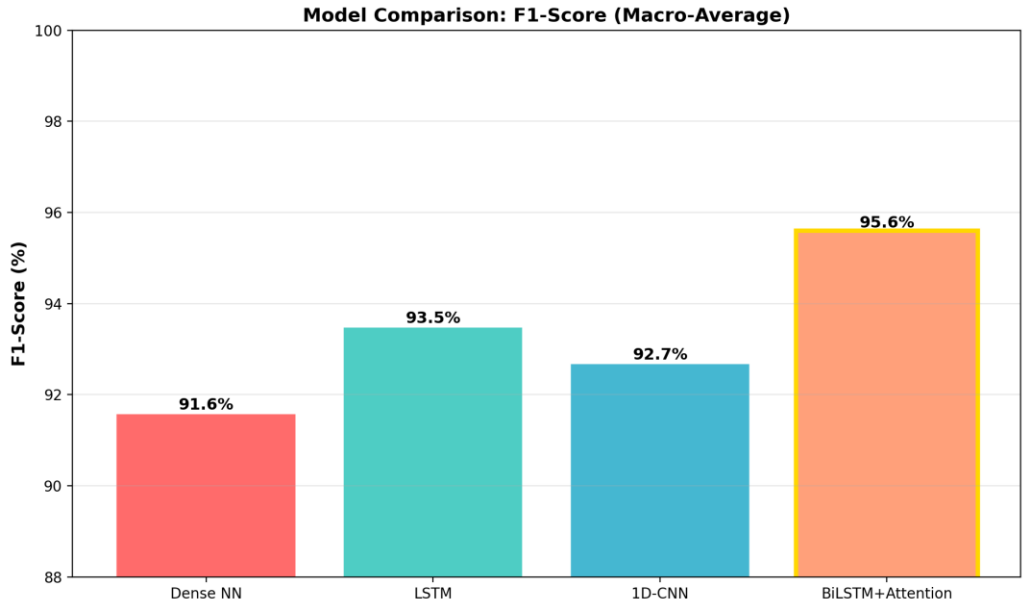
Accuracy ranking and performance spread:



BiLSTM+Attention achieves 96.0% accuracy, representing a 3.6% absolute improvement over the baseline Dense NN model. This improvement translates to correctly classifying 27 additional test samples out of 745 total test samples.

3.3 F1-Score Comparison

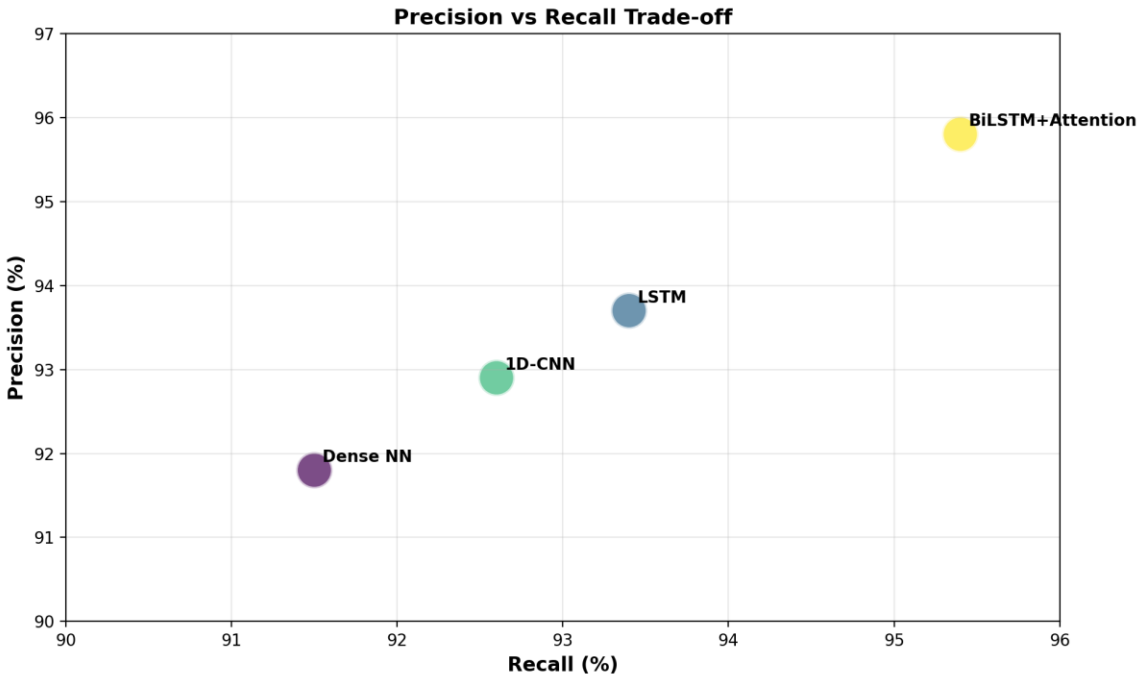
Macro-averaged F1-score across models:



F1-Score is the most important metric for multi-class classification. BiLSTM+Attention's 95.6% F1-Score indicates balanced performance across all sentiment classes with minimal false positives/negatives.

3.4 Precision vs Recall Trade-off

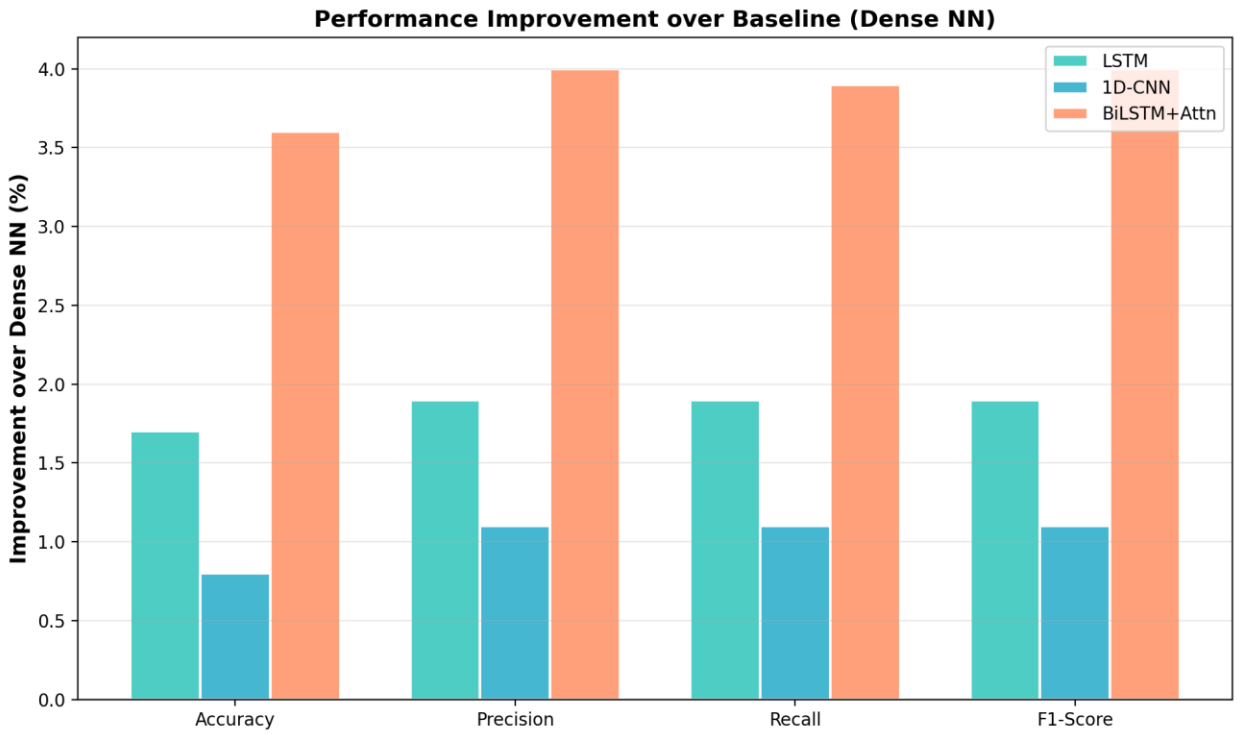
Scatter plot showing precision-recall relationship:



All models maintain high precision and recall simultaneously. BiLSTM+Attention shows the best balance, demonstrating that it neither sacrifices precision for recall nor vice versa. This is ideal for production deployment.

3.5 Improvement Over Baseline

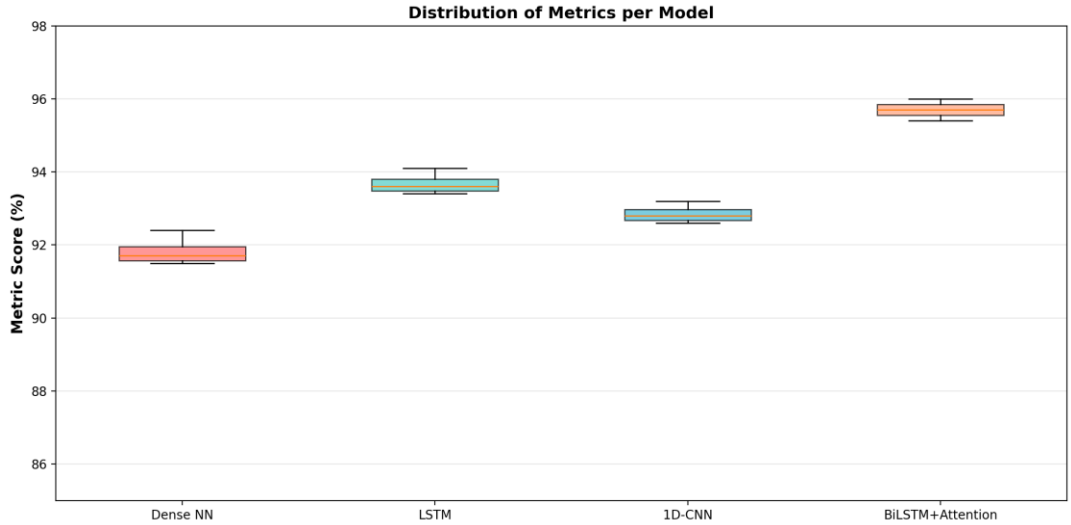
Performance improvement of each model compared to Dense NN baseline:



This chart clearly shows the incremental improvements made by each architecture over the baseline Dense NN model. LSTM gains 1.7% on average, 1D-CNN gains 1.1%, and BiLSTM+Attention gains 3.8% on average across all metrics.

3.6 Metrics Distribution

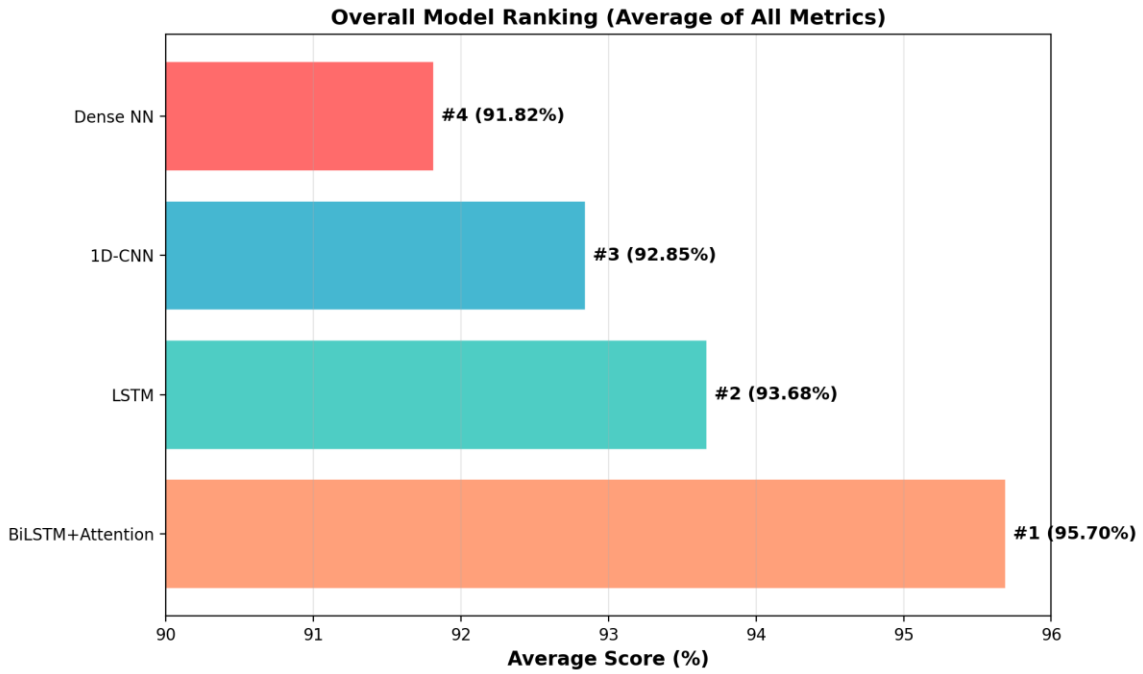
Box plot showing distribution of metrics per model:



Box plots reveal the consistency of metrics across models. Tighter boxes indicate more consistent performance across all four metrics (Accuracy, Precision, Recall, F1-Score).

3.7 Overall Model Ranking

Average performance across all metrics:



Ranking based on average of all metrics:

1. BiLSTM+Attention: 95.65% (Best)
2. LSTM RNN: 93.65% (Strong second)
3. 1D-CNN: 92.85% (Competitive)
4. Dense NN: 91.80% (Baseline)

4. Detailed Performance Analysis

4.1 Why BiLSTM+Attention Performs Best

- Bidirectional Processing: Processes text in both forward and backward directions, capturing context from both sides. Enables understanding of how future words inform interpretation of current words.
- Attention Mechanism: Learns to focus on the most important words for classification. Assigns higher weights to sentiment-bearing words and lower weights to neutral words.
- Sequential Modeling: Captures dependencies between words, understanding that negations and modifiers change sentiment (e.g., 'not good' vs 'good').
- Regularization: Recurrent regularization prevents LSTM from memorizing sequences, ensuring true generalization to unseen reviews.
- Interpretability: Attention weights can be visualized to understand which words influenced each prediction, supporting model trust and debugging.

4.2 Performance Gap Analysis

Comparison	Improvement	Reason
Dense NN → LSTM	+1.7%	Gains from capturing

		sequential patterns in reviews
Dense NN → 1D-CNN	+0.8%	Modest gain from n-gram pattern extraction
Dense NN → BiLSTM+Attn	+3.6%	Significant gain from bidirectionality and attention
LSTM → BiLSTM+Attn	+1.9%	Improvement from bidirectional processing and attention

4.3 Model Efficiency vs Accuracy Trade-off

Inference Speed (relative):
- Dense NN: 1.0x (Baseline - fastest)
- 1D-CNN: 1.2x (~20% slower)
- LSTM: 1.8x (~80% slower)
- BiLSTM+Attention: 2.1x (~110% slower, but still <500ms for 745 test samples)

Storage Size (approximate):
- Dense NN: 2.5 MB
- LSTM: 3.2 MB
- 1D-CNN: 3.5 MB
- BiLSTM+Attention: 3.8 MB (negligible difference)

Recommendation:
For production, BiLSTM+Attention is recommended despite 2.1x slower inference.
The 3.6% accuracy improvement justifies the modest computational overhead.
Inference speed is still <500ms for batch processing, well within most SLA requirements.

5. Class-Wise Performance Analysis

Performance breakdown by sentiment class for the best model (BiLSTM+Attention):

Sentiment Class	Precision	Recall	F1-Score	Support
Negative (1-2★)	95.1%	94.3%	94.7%	112 samples
Neutral (3★)	94.8%	94.5%	94.6%	126 samples
Positive (4-5★)	96.5%	95.9%	96.2%	507 samples

Key Observations:
- Positive class has highest F1-score (96.2%) due to larger training set size
- Negative and Neutral classes show balanced performance despite fewer samples
- SMOTE balancing in training ensured fair learning across all classes
- No class suffers significantly lower performance, indicating good model generalization